Tumor–stroma coupling identifies a reproducible prognostic program in pancreatic cancer

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Abstract

Pancreatic ductal adenocarcinoma (PDAC) outcomes reflect coordinated tumor-intrinsic and stromal states, but bulk transcriptomic methods typically decompose these compartments without survival supervision. We developed DeSurv, a survival-supervised matrix factorization framework that jointly learns gene programs and survival associations. In TCGA and CPTAC PDAC, DeSurv identified a compact three-program structure organized along a co-varying tumor–stroma axis, with Classical tumor/restCAF-like stroma at one pole and Basal-like tumor/proCAF-like stroma at the high-risk pole. Across five external cohorts, each standard-deviation shift toward this pole was associated with a pooled hazard ratio of 1.50 (95% CI, 1.31–1.72) and remained prognostic after adjustment for PurIST and DeCAF; dichotomized high-risk patients had a hazard ratio of 1.79 (95% CI, 1.41–2.28). Projected without retraining into treated cohorts, the same programs revealed therapy-dependent remodeling, with the proCAF-like program becoming protective. Together, these results identify tumor–stroma coupling as a reproducible prognostic program in PDAC.

Pancreatic ductal adenocarcinoma (PDAC) is among the deadliest solid tumors, with a five-year relative survival rate of approximately 13%[1] and limited responsiveness to existing therapies. Bulk transcriptomic studies have defined reproducible malignant programs, most consistently Basal-like and Classical tumor states[2–4], as well as stromal programs including tumor-promoting proCAF and tumor-restraining restCAF states[5]. Single-cell and spatial studies further show that PDAC tumors comprise malignant, stromal, immune and other microenvironmental components whose transcriptional states can co-vary within the same lesion[6,7]. Whether joint configurations of these programs carry prognostic information beyond discrete tumor and stromal labels, and whether such configurations can be recovered de novo from bulk transcriptomes, remains incompletely understood.

Bulk RNA profiling remains the primary platform for cohort-scale survival annotation and prognostic modeling in large clinical cohorts[8,9], but each profile is a composite of malignant cells, cancer-associated fibroblasts, immune infiltrates and other cellular sources[10]. Nonnegative matrix factorization (NMF) has been instrumental for resolving this mixture into additive, interpretable gene programs[3,11–13]. In PDAC, this strategy has yielded major biological insights, including molecular subtypes[2,4,5,14], virtual microdissection of Basal-like and Classical tumor programs[3], and compartment-specific deconvolution of tumor and microenvironmental signals[13].

A limitation of this workflow is that programs are usually discovered without outcome information and tested for clinical association only after factorization[2,14]. Because unsupervised NMF minimizes reconstruction error, its factors tend to capture the highest-variance structure in the data, which may reflect tissue composition or other outcome-neutral variation rather than prognostic biology[8,15]. In PDAC, exocrine and other low-purity signals can dominate unsupervised factors while carrying limited prognostic value[4,5]. More broadly, tumor purity accounts for a large fraction of expression variation across cancer types[16], illustrating how reconstruction-optimal and outcome-relevant subspaces can diverge. Reconstruction alone also leaves the factor basis non-identifiable[17–19], contributing to cross-study variability among unsupervised cancer subtype taxonomies[20,21].

This raises a biological and methodological question: which bulk-transcriptomic programs carry reproducible prognostic information in PDAC, and how are those programs distributed across malignant, stromal, immune and other components of the tumor microenvironment? Neither extreme suffices on its own. Unsupervised factorization optimizes reconstruction and can over-capture dominant composition-driven variation, whereas selecting genes directly on survival can identify outcome-associated expression features but leaves their cellular source ambiguous. A gene's prognostic association may reflect malignant, stromal, immune or other microenvironmental variation.

Here we present DeSurv, a survival-supervised deconvolution framework that integrates NMF with Cox proportional hazards modeling so that gene programs are shaped during discovery by both reconstruction and survival. Supervision is applied to the gene-program matrix while preserving the mixture-coefficient interpretation of sample loadings, and trained programs score new samples by projection without retraining. We evaluate DeSurv through two non-circular tests: recovery of ground-truth program identities in simulations where the true gene sets are known a priori, and out-of-sample generalization to five independent PDAC cohorts on which no model parameters were estimated.

In simulations, DeSurv recovers prognostic programs more reliably than standard NMF, with advantages that scale with the separation between high-variance and prognostic subspaces and vanish when no survival signal is present. In PDAC, DeSurv recovers a compact three-program structure organized around a co-varying tumor–stroma axis linking Classical tumor identity and restCAF-like stroma at one pole with Basal-like tumor identity and proCAF-like stroma at the high-risk pole. This score generalizes across five independent PDAC cohorts without retraining and remains prognostic beyond established PurIST and DeCAF classifications. These findings identify tumor–stroma coupling as a reproducible prognostic program in PDAC, recoverable de novo from bulk transcriptomes by survival-supervised matrix factorization.

Results

Model Overview

We developed DeSurv, a survival-supervised deconvolution framework that learns two linked quantities from bulk transcriptomes paired with patient outcomes: gene programs and their survival associations (Fig. 1A, Methods). DeSurv combines nonnegative matrix factorization with Cox proportional hazards modeling so that each gene program is shaped by both reconstruction of the expression matrix and association with survival. The survival objective acts on the gene programs but not on the sample loadings, allowing the learned programs to be applied to new samples by projection without retraining. A supervision parameter controls the balance between reconstruction and survival, and both this parameter and the factorization rank are selected by Bayesian optimization on cross-validated concordance (Fig. 1B, Methods).

Applied to the PDAC training cohorts from TCGA[22] and CPTAC[23] (Materials and Methods), cross-validated concordance varied across candidate ranks but plateaued from $k = 3$ through $k = 12$ (SI Appendix, Fig. S4), indicating that three supervised factors capture nearly as much prognostic information as twelve. Within this plateau, joint Bayesian optimization over rank and supervision strength with a one-standard-error rule (Methods) accordingly selected the parsimonious $k = 3$ at $\alpha = 0.334$. Peak cross-validated C-index was 0.655 across $n = 273$ patients with 139 events. Standard NMF rank-selection diagnostics (reconstruction residuals, cophenetic correlation, silhouette width)[12,24] did not converge on a consistent rank in PDAC (SI Appendix, Fig. S5), motivating the supervised approach.

Survival supervision and standard NMF produce different factor structures in PDAC

To examine how survival supervision shapes the recovered factor structure, we fit standard NMF at the same rank as DeSurv ($k = 3$, selected above) and examined the overlap between each method's factor-specific gene rankings and established PDAC gene programs (Fig. 2A–B). Matching k across methods isolates the effect of survival supervision on factor content from differences in model complexity; a sensitivity analysis at NMF's independently selected ranks (elbow $k = 5$, BO $k = 7$) is provided in SI Appendix, Table S4. We first asked which malignant, stromal, immune and other microenvironmental programs were represented in each factorization, and then tested whether these structural differences corresponded to differences in survival contribution.

We refer to the three DeSurv factors as D1–D3 and the three NMF factors as N1–N3. In the DeSurv factorization, D1 correlated with Classical tumor programs and restCAF-like stromal signatures, D2 with proCAF-like programs, and D3 with Basal-like tumor programs (Fig. 2A). Among the programs detected, DeSurv organized the dominant survival-associated variation along a tumor–stroma axis, with D1 representing the Classical/restCAF-like low-risk pole and increasing risk reflecting movement toward Basal-like and proCAF-like structure, consistent with evidence that survival stratification in PDAC improves when tumor-intrinsic and CAF subtypes are considered jointly[5,25]. No DeSurv factor showed a significant positive correlation with exocrine-associated gene programs (DECODER Exocrine, Collisson Exocrine, Bailey aberrantly differentiated endocrine–exocrine (ADEX); Fig. 2A). The survival contributions of these factors and their relationship to reconstruction of the expression matrix are quantified below.

Standard NMF produced a different factor organization (Fig. 2B). One factor (N2) was strongly correlated with exocrine-associated gene programs and negatively correlated with immune and stromal signatures (Fig. 2B). A second factor (N1) correlated with malignant-epithelial (tumor) identity, carrying both Classical and Basal-like markers (Fig. 2B). A third factor (N3) correlated with immune, fibroblast, and extracellular matrix (ECM)-related programs (including Puleo Desmoplastic, SCISSORS proCAF and restCAF[26], Maurer Immune and ECM[27], and DECODER Activated Stroma), aggregating these into a single factor rather than separating them as DeSurv does (D1 and D2; Fig. 2A). At $k = 3$, NMF thus devotes one of its three factors to exocrine-associated variation, leaving two factors to capture all tumor-intrinsic and microenvironmental programs. Whether these structural differences correspond to differences in prognostic information is quantified next.

NMF seeks a low-rank *reconstruction* of the transcriptome; many factorizations reconstruct comparably well, and the unsupervised objective gives no incentive to prefer a prognostically informative direction. We therefore quantified per-factor contribution using two metrics (Fig. 2C; SI Appendix, Sections 10–11): each factor's contribution to reconstruction (the normalized reconstruction Shapley value, which partitions the model's reconstruction and sums to 100%) and its survival contribution (the change in Cox partial log-likelihood $\Delta\ell$ when that factor is dropped from a k-factor Cox model). NMF distributed reconstruction across its factors in proportions unrelated to prognosis: all three NMF factors had negligible $\Delta\ell$ despite spanning a wide range of reconstruction share, while DeSurv concentrated nearly all survival contribution into D1. Specifically, D1 contributed 18.8% of reconstruction with $\Delta\ell = 66.4$, while the largest-reconstruction NMF factor (N1) contributed 41.8% of reconstruction but only $\Delta\ell = 0.1$ to survival. The exocrine-associated NMF factor (N2) contributed 18.6% of reconstruction yet only $\Delta\ell = 1.16$ to survival. The largest-reconstruction DeSurv factor (D3, 56% of reconstruction) likewise contributed minimally to survival after accounting for D1. Thus, in PDAC the factors that contribute most to reconstruction are not the most prognostic, consistent with tumor purity analyses across cancer types[16].

To further characterize how the two factorizations relate, we examined pairwise W-matrix gene loading correlations over all 1,970 genes (Fig. 2D; Spearman over the full loading profile, since a top-gene subset would overstate preservation). At $k = 3$, NMF captures the entire malignant-epithelial compartment in a single factor (N1), which we therefore label "tumor' ' rather than Classical; N1 carries both Classical and Basal markers and maps strongly onto DeSurv's Basal-like tumor factor (N1$\leftrightarrow$D3 Spearman 0.84), a tumor$\leftrightarrow$tumor correspondence, while NMF's microenvironmental factor maps onto DeSurv's proCAF stroma (N3$\leftrightarrow$D2 0.87). DeSurv's most prognostic factor (D1) is essentially uncorrelated with all three NMF factors (entire D1 column ≤ 0.36): because D1 couples Classical tumor and restCAF-like stromal signatures into a single survival-aligned program, its structure is novel relative to NMF rather than a reweighting of an existing factor (SI Appendix, Table S2). NMF's exocrine factor (N2) has no strong DeSurv counterpart (row capped at 0.61), consistent with its dissolution under supervision. In short, NMF lumps the tumor compartment into one factor, whereas DeSurv resolves it into Classical- (D1) and Basal-aligned (D3) programs and constructs the survival-aligned tumor–stroma coupling (D1) that NMF does not isolate. Whether this translates to improved generalization across independent cohorts is tested below.

Survival-aligned programs generalize across independent PDAC cohorts with lower complexity

A prognostic factorization is useful only if it generalizes beyond the training cohort. Having established that survival supervision and standard NMF produce different factor structures in the training data, we next tested whether each method's structure transfers to independent cohorts without retraining. The five external validation cohorts (Dijk, Moffitt, PACA-AU array and RNA-seq, and Puleo) likewise consist of non-metastatic, treatment-naive specimens (SI Appendix, Table S1). We computed factor scores in each validation dataset by projecting the gene programs learned in the training cohort ($Z = \widetilde{W}^{\top} X_{\text{new}}$), yielding a continuous score for each factor per sample.

External validation C-index for NMF at independently selected ranks is summarized in SI Appendix, Table S4. Standard NMF at elbow-selected $k = 5$ and BO-selected $k = 7$ substantially improved over NMF at $k = 3$, with $k = 7$ matching or exceeding DeSurv's per-cohort C-index in most cohorts, though only when DeSurv's tuning framework is applied ($\alpha = 0$ in the joint BO over rank and hyperparameters), since NMF's own rank-selection heuristics (cophenetic, silhouette, residual; SI Appendix, Fig. S5) do not converge on this rank. However, this concordance gain still requires more than twice as many factors. DeSurv at $k = 3$ retains a significant adjusted hazard ratio after adjustment for PurIST and DeCAF (per-SD HR = 1.33, 95% CI 1.13–1.56, $P < 0.001$; Table 1), achieving independent prognostic value more parsimoniously. When β coefficients are re-fit on each validation cohort, DeSurv's three-factor decomposition is BIC-preferred over NMF's seven-factor decomposition in three of five cohorts, with the remaining two essentially tied (SI Appendix, Table S5). Across 5 independent PDAC cohorts ($n = 616$), the DeSurv linear predictor showed a consistent survival association (stratified Cox model: pooled HR per SD 1.50; 95% CI 1.31–1.72; $P = 2.6 \times 10^{-9}$; per-cohort forest plot in Fig. 3A; KM curves in SI Appendix, Fig. S6C–F). At the matched rank ($k = 3$), the analogous NMF linear predictor showed a weaker pooled effect (HR per SD 1.11; 95% CI 1.00–1.23; $P = 0.057$); a direct comparison using dichotomized risk groups is presented below.

To assess clinical interpretability of these continuous associations, we dichotomized validation samples into high- and low-risk groups using method-specific cutpoints, each independently optimized via cross-validation on the training data. The DeSurv linear predictor separated survival trajectories in the pooled validation cohort (HR = 1.79, 95% CI 1.41–2.28, $P < 0.001$; Fig. 3B). The same procedure applied to NMF at $k = 3$ yielded weaker stratification (HR = 1.38, 95% CI 0.92–2.09, $P = 0.122$; Fig. 3C). The DeSurv cutpoint classified a larger validation subset as high risk (151 vs 37 high-risk patients; 114 reclassified) while preserving strong survival separation. The DeSurv high-risk group was enriched for both established poor-prognosis subtypes (Basal-like by PurIST, proCAF by DeCAF; Fisher's exact, SI Appendix, Fig. S7), capturing established consensus biology while adding independent prognostic information. The same analysis applied to NMF risk groups at $k = 3$ showed Basal-like enrichment but no significant proCAF enrichment (SI Appendix, Fig. S7), consistent with NMF's failure to isolate the tumor–stroma coupling that underlies D1. Together, these results indicate that survival supervision yields a compact, biologically interpretable prognostic structure that generalizes across independent PDAC cohorts; DeSurv identifies a lower-dimensional tumor–stroma coupling program with independent prognostic value beyond PurIST and DeCAF, whereas higher-rank NMF models can achieve similar per-cohort concordance only at substantially greater complexity without recovering the same coupling axis.

Table 1: Pooled external validation hazard ratios (per SD of the linear predictor) for DeSurv at $k = 3$ and standard NMF at BO-selected $k = 7$ ($\alpha = 0$), across the five validation cohorts ($n = 616$ patients, 414 events). Adjusted models include the PurIST tumor-intrinsic and DeCAF stromal molecular classifiers as covariates, with stratification by dataset. Both methods retain significant prognostic value after adjustment, with DeSurv's larger HR drop reflecting biological concordance with established subtypes and NMF $k = 7$'s smaller drop reflecting orthogonal residual content from its additional, less interpretable factors.

Method	Unadjusted HR per SD (95% CI), P	Adjusted HR per SD (95% CI), P
DeSurv ($k = 3$)	1.50 (1.31–1.72), < 0.001	1.33 (1.13–1.56), < 0.001
Std NMF ($k = 7$, BO $\alpha = 0$)	1.48 (1.34–1.64), < 0.001	1.47 (1.27–1.70), < 0.001

To establish that the dominant program is more than a continuous restatement of the discrete subtype calls, we partitioned D1 against both classifiers across the pooled validation cohorts ($n = 616$, 414 events). The binary PurIST and DeCAF calls together accounted for 28% of the within-cohort variance of the continuous D1 score, indicating that D1 grades tumour–stroma state more finely than the dichotomous subtypes it aligns with. More importantly, D1 retained significant prognostic value after conditioning on both classifiers (adjusted HR per SD 0.82, 95% CI 0.74–0.92, $P = 0.0006$; partial likelihood-ratio $\chi^2 = 11.6$ on 1 df, $P = 0.0006$), improving the stratified-Cox concordance over a PurIST-plus-DeCAF model (C-index 0.593 to 0.621). The coupling axis therefore carries prognostic information that neither the tumour-intrinsic nor the stromal classifier captures alone. The axis is also not specific to DeSurv's optimizer: three independent survival-supervised methods (supervised principal components, Cox partial least squares, and sparse Cox) recover the same D1 score in the training cohort (mean $|r| = 0.83$), whereas the leading unsupervised direction does not ($|r| = 0.10$), identifying D1 as a property of survival supervision rather than of the algorithm (SI Appendix, supervised-method recovery).

The recovered programs reveal a context-dependent prognostic shift in treated PDAC

Because DeSurv scores new samples by projecting the trained gene-program matrix without re-estimating any parameters, the same programs can be read directly in treated cohorts, allowing PDAC's prognostic architecture to be compared across disease settings. We projected the trained programs onto three bulk treated cohorts — the metastatic gemcitabine-backbone trials Rash ($n = 85$) and Accept ($n = 70$) and the borderline-resectable/locally-advanced Linehan cohort (FOLFIRINOX $\pm$ CCR2 inhibitor, $n = 66$) — and, because these settings differ in stage and regimen, analysed each separately rather than as a pooled estimate (Fig. 4; SI Appendix, treated-cohort analysis). Because the trained survival coefficient for the proCAF program (D2) was zero, programs were evaluated per factor.

The proCAF stromal program D2 was associated with longer overall survival in every treated cohort (PurIST + DeCAF-adjusted HR per SD 0.46 (95% CI 0.31–0.70, $P < 0.001$) in Linehan, 0.73 (95% CI 0.52–1.04, $P = 0.078$) in Accept, and 0.77 (95% CI 0.53–1.12, $P = 0.178$) in Rash) — a consistent protective direction across three independent settings and chemotherapy backbones — significant in the borderline-resectable cohort (Linehan) and non-significant but directionally concordant in the metastatic cohorts (Accept and Rash). Because D2 is a stromal program and is not collinear with the tumour-intrinsic and CAF classifiers, its association was not an artifact of adjustment: where it reached significance (Linehan), the adjusted effect strengthened rather than attenuated, and across all three cohorts the direction was consistently protective. D2 was also associated with longer progression-free survival in the metastatic cohort and increased from pre- to post-treatment in patients with paired biopsies, indicating that the proCAF program is itself remodelled by therapy (SI Appendix). The cohorts were analysed separately rather than pooled; the consistent protective direction across three independent cohorts, and near-identical results under a full-gene projection (Spearman $r \geq 0.86$ between the top-gene and full-gene scores; SI Appendix), provide the cross-cohort evidence.

By contrast, the basal–classical program D1 showed a setting-dependent association that tracked the basal–classical axis itself rather than any property unique to D1 (it separated PurIST calls at AUC 0.84–0.89 in every cohort). D1 remained prognostic in the less-advanced Linehan cohort (marginal HR 0.58 (95% CI 0.36–0.94, $P = 0.027$)) but attenuated in the metastatic cohorts — as did every established basal–classical classifier (PurIST, Moffitt, and Bailey were all non-significant; SI Appendix), indicating a stage-dependent attenuation of the basal–classical prognostic axis rather than a failure of D1: Basal-like prevalence in the metastatic cohorts was the expected 16–22%, and the same axis is prognostic at $P < 10^{-8}$ in equally-prevalent treatment-naive cohorts (the Linehan estimate rests on only six Basal-like cases and is correspondingly imprecise). We therefore report D1 marginally, because adjusting it for the collinear PurIST classifier conditions the same axis on itself. D3 (Basal-like) was not associated with overall survival in any treated cohort (consistent with the basal–classical contrast residing in D1) but was associated with shorter progression-free survival in the metastatic cohort (HR per SD 1.35 (95% CI 1.08–1.69, $P = 0.009$)), consistent with the aggressive biology of Basal-like tumours; the D1 score independently validated against protein-level GATA6 in-situ hybridisation (Spearman $\rho = 0.68$, $P < 0.001$; SI Appendix). Because treatment was confounded with cohort and stage, these analyses address prognosis rather than prediction.

DeSurv recovers prognostic gene programs when variance and prognosis diverge

The cross-cohort generalization above is corroborated by simulations with known ground-truth latent structure. To test whether survival supervision improves gene program recovery under nonnegative constraints, we designed simulations in which prognostic gene programs explained low variance relative to outcome-neutral background signals, with survival times generated from latent factor scores via a Cox model ($p = 3{,}000$ genes, $n = 200$ samples, true $k = 3$; Materials and Methods, Simulation Studies). To test whether survival information should enter during factorization rather than only afterward, we compared DeSurv to standard NMF ($\alpha = 0$), where factors are learned by reconstruction error alone and survival associations are estimated by a separately fit Cox model. Both methods were tuned via BO over cross-validated C-index, so that differences in performance reflect the role of supervision during factorization, not during model selection.

In the primary scenario, where prognostic programs explained low variance relative to outcome-neutral programs, DeSurv consistently achieved higher C-index (Fig. 5A) and substantially higher precision for recovering the true prognostic gene programs (Fig. 5B). Here precision denotes the fraction of a factor's top-weighted genes (those with the largest factor specificity score s_{ij}, where n_{top}, the number of factor-specific genes entering the survival linear predictor, is chosen by BO; SI Appendix, Sections 6 and 9) that overlap with a true prognostic program's gene set. The unsupervised baseline exhibited near-zero precision, indicating that variance-driven factorization failed to concentrate on the genes that actually drove survival. This confirms that outcome supervision during factorization is necessary to recover prognostic programs that do not dominate expression variance. Across 100 replicates, DeSurv achieved median C-index 0.837 versus 0.724 for standard NMF ($\Delta = 0.113$, paired Wilcoxon $P < 0.001$). The precision advantage exceeded the C-index advantage, reflecting that unsupervised factorization can match predictive performance while assigning different genes to each factor; survival supervision constrains gene-program identity itself. A similar decoupling appears empirically in PDAC at $k = 5$ (NMF's elbow-selected rank), where supervised (DeSurv) and unsupervised (NMF) factorizations yield substantially different factor-specific top genes (SI Appendix, Fig. S8 [DeSurv $k = 5$] vs Fig. S9 [NMF $k = 5$]), with similar C-indices in external cohorts.

To verify that these gains reflect genuine signal recovery rather than overfitting, we repeated the analysis under a null scenario ($\beta = 0$) and a mixed scenario with partial overlap between variance-dominant and prognostic structure (SI Appendix, Fig. S3). DeSurv's benefit scaled with the degree to which prognostic programs were masked by outcome-neutral variation: largest in the primary scenario ($\Delta = 0.113$), attenuated when variance and prognosis partially overlapped ($\Delta = 0.069$, paired Wilcoxon $P < 0.001$), and absent when no survival signal existed, where both methods yielded C-index near 0.5. DeSurv also recovered the true rank ($k = 3$) more frequently than standard NMF, which systematically under-selected near $k = 2$ (Fig. 5C).

Discussion

PDAC's prognostic transcriptional structure, as recovered by survival-supervised factorization, is dominated by a single axis that couples Classical tumor identity with restCAF stroma rather than by tumor-intrinsic and stromal programs isolated separately. This coupling generalizes across five independent PDAC cohorts without retraining and remains significant after adjustment for PurIST and DeCAF, suggesting that the prognostically relevant biology in PDAC is the joint configuration of tumor and stromal compartments rather than either alone. The framework that recovered this structure, DeSurv, concentrates prognostic signal into fewer, more interpretable gene programs by coupling NMF to a Cox survival objective during discovery; its architecture (supervision on W, mixture interpretation preserved on H) enables single-sample scoring by projection, a transportability property not shared by methods that route supervision through H[28,29].

In PDAC, the factor structure produced by survival supervision is biologically informative. The factor with the largest survival contribution (D1) couples Classical tumor programs with restCAF stromal signatures, consistent with growing evidence that clinical outcomes depend on the joint configuration of tumor-intrinsic and cancer-associated fibroblast subtypes[5,25]. This co-occurrence within a single factor suggests that the prognostically relevant biology in PDAC is not purely tumor-intrinsic but reflects tumor–stroma coupling that unsupervised methods distribute across multiple factors. Standard NMF, by contrast, allocated a dedicated factor to exocrine-compositional variation, a signal that reflects tumor purity rather than cancer biology[4], leaving less capacity for the microenvironmental programs that DeSurv separates at the matched rank ($k = 3$). The contribution of this work is twofold. Methodologically, these programs are recovered de novo, without pre-specified signatures, with direct quantification of their survival associations during factorization. Biologically, this recovery surfaces tumor–stroma coupling as a single survival-aligned program — a relationship implicit in prior virtual microdissection[3], experimental microdissection[27], and unsupervised deconvolution[13] but not previously isolated as a single bulk-tx program.

Survival supervision concentrates prognostic signal into fewer factors, creating a broad concordance plateau that the one-standard-error rule leverages to select parsimonious ranks, a deliberate balance between biological completeness and clinical relevance. In PDAC, this plateau spanned $k = 3$–12 (SI Appendix, Fig. S4); for standard NMF, concordance increased steadily with k, approaching DeSurv's level by $k = 7$ (SI Appendix, Table S4) but requiring more than twice as many factors and producing a fragmented structure. Notably, NMF's own unsupervised rank selection heuristics (reconstruction residuals, cophenetic correlation, and silhouette width; SI Appendix, Fig. S5) do not point to $k = 7$; the elbow criterion suggests $k = 5$, which does not match DeSurv's concordance. NMF only reaches $k = 7$ through Bayesian optimization with $\alpha = 0$, that is, by applying DeSurv's tuning infrastructure to an unsupervised model. Beyond this parsimony, DeSurv at $k = 3$ achieves independent prognostic value: both methods retain significance after classifier adjustment (Table 1). The model selection framework (joint BO over rank and hyperparameters with the 1-SE rule) is itself part of the methodological contribution, not merely the survival gradient: parsimony is a property of the 1-SE rule, but supervision creates the flat concordance surface under which the rule can operate effectively.

Our simulations provide further guidance on when this tradeoff is most beneficial: DeSurv's advantage is greatest when prognostic programs explain modest variance relative to outcome-neutral signals and vanishes under null conditions where no survival signal exists. DeSurv is therefore not intended to replace unsupervised NMF in all settings. When the goal is exploratory discovery without clinical endpoints, when survival data is sparse or unreliable, or when the application requires comprehensive biological characterization rather than prognostic stratification, unsupervised methods remain appropriate; recent reviews of cancer transcriptomic deconvolution methods provide broader guidance on method selection across these settings[30]. The α parameter, tuned via Bayesian optimization, lets the data determine the appropriate balance for a given cancer type and cohort size, and the $\alpha = 0$ endpoint recovers standard NMF as a special case. DeSurv is likewise not proposed as a competing molecular classifier; translating gene programs into discrete clinical subtype labels would be a downstream application. The relevant comparison with existing classifiers is whether DeSurv's factors capture independent prognostic signal beyond what those classifiers already explain; Table 1 confirms this directly, with the DeSurv $k = 3$ linear predictor retaining a significant adjusted hazard ratio after conditioning on PurIST and DeCAF (HR = 1.33, 95% CI 1.13–1.56, $P < 0.001$).

Several limitations define the scope of these findings. First, DeSurv assumes Cox proportional hazards, which may not hold in settings with delayed or non-proportional treatment effects; the convergence guarantee (SI Appendix, Theorem 1) applies to an idealized algorithm, and the implementation includes practical modifications that depart from the theoretical analysis, though empirically these do not affect convergence behavior. Extensions to more flexible survival models are a natural direction. Second, DeSurv was applied here only to PDAC, and broader benchmarking across cancer types is needed to establish the generality of the gap between reconstruction-optimal and survival-optimal subspaces; the computational cost of Bayesian optimization may limit scalability to very large cohorts, though hyperparameter selection is performed once and the final model applies by projection. Third, the cohorts used to train the factorization and for primary out-of-sample validation consist of non-metastatic, treatment-naive PDAC specimens; the model is therefore learned in resectable disease. Applying the same programs by projection to treated cohorts revealed that PDAC's prognostic architecture is context-dependent. The proCAF stromal program D2 — prognostically silent at baseline, with a trained survival coefficient of zero — was protective on overall survival in all three treated cohorts (and on progression-free survival in the metastatic series) and increased from pre- to post-treatment in patients with paired biopsies, whereas the basal–classical tumour axis (D1) was prognostic in less-advanced disease but attenuated in metastatic disease in step with the established PurIST, Moffitt, and Bailey classifiers (all non-significant in the metastatic cohorts), indicating a stage-dependent property of the basal–classical axis rather than a failure of

D1 (Results; SI Appendix, treated-cohort analysis). These two axes thus show reciprocal, externally-corroborated context-dependence — the tumour program dominating prognosis in resectable disease and the stromal program emerging under treatment — consistent with growing evidence that PDAC stroma is functionally dual and that discrete restraining fibroblast states limit progression under treatment[5,31–33]. These treated-cohort associations are prognostic rather than predictive, because treatment was confounded with cohort and stage and biopsies were predominantly pre-treatment; establishing whether the proCAF program identifies patients who benefit from stroma- or myeloid-directed therapy — a hypothesis supported by its CD163$^+$ macrophage content and by the observation that proCAF-dominant tumours preferentially respond to myeloid inhibition[5] — will require randomized trials with the program measured prospectively, ideally in the neoadjuvant setting where treated and resectable disease coincide and the trained gene-program matrix can be scored by projection without retraining. Fourth, the tumour–stroma coupling is defined here at the level of bulk gene programs — the covariation of Classical-tumour and restCAF-stromal signatures across patients — and not as per-cell co-expression or spatial adjacency, which are distinct and stronger hypotheses that single-cell and spatial assays, rather than the present bulk model, are positioned to test. That this covariation is not an artefact of tumour purity — the prerequisite for its bulk-level interpretation — is established directly: D1 remains prognostic after adjustment for PurIST and DeCAF and is recovered by independent survival-supervised methods but not by the leading unsupervised direction (Results). Resolving the coupling at cellular or spatial resolution, which would require deconvolution-based scoring rather than transfer of the bulk gene-program matrix to those sparse substrates, is a valuable future direction rather than a validation on which this bulk finding depends. Bulk profiling continues to underpin clinical molecular stratification, from trial programs such as COMPASS to CLIA-validated diagnostic assays in routine clinical care, a deployment landscape favored by the cost, FFPE compatibility, and standardization requirements of the clinical workflow. As single-cell cohorts with clinical annotation grow, DeSurv's gene programs, which align with cell-type signatures confirmed by independent single-cell studies[6,7], can serve as a bridge between bulk-derived prognostic models and emerging cellular-resolution datasets.

The standard workflow in cancer transcriptomics (unsupervised discovery followed by retrospective evaluation) incurs a structural cost whenever the highest-variance signals are not the most prognostic, likely the norm in cancer types where tumor purity and tissue composition dominate expression variation[16]. DeSurv addresses this by aligning factorization with the clinical question from the outset: in PDAC, this recovered a Classical-tumor + restCAF program at a parsimonious $k = 3$, and the resulting DeSurv predictor transferred to five independent cohorts (pooled HR per SD 1.50) and survived PurIST/DeCAF adjustment. The supervised objective additionally pins the gene-program basis, addressing the basis-non-identifiability of reconstruction-only NMF: rotations of the factor matrix paired with compensating Cox coefficients can preserve overall predictive performance while shifting which genes load most strongly on each factor, so unsupervised gene programs need not reproduce across heterogeneous cohorts. This non-identifiability contributes to the cross-study variability observed among prior unsupervised PDAC subtype taxonomies[2,3,14]. Although a purely supervised predictor fit on bulk expression can match this external prediction, it returns a single, compartment-ambiguous risk score rather than source-resolved programs, so predictive performance alone does not establish that a signature reflects a specific cellular program; by retaining the reconstruction term, DeSurv instead keeps its programs resolved at the gene-program level—D1's loadings span Classical-tumour and restCAF-stromal genes, concordant with established compartment-specific signatures from virtual and experimental microdissection[3,27] and CAF deconvolution[5]—while supervision aligns them with outcome. The combination of a survival-supervised gradient and joint Bayesian optimization with the one-standard-error rule provides a principled framework for extracting outcome-relevant programs from bulk tumor transcriptomes wherever clinical outcomes are available; as clinically annotated cohorts accumulate, this setting is more likely to expand alongside than be displaced by single-cell technologies.

Methods

Problem formulation and notation

Let $X \in \mathbb{R}_{\geq 0}^{p \times n}$ denote the nonnegative gene expression matrix. DeSurv approximates $X \approx WH$, where $W \in \mathbb{R}_{\geq 0}^{p \times k}$ contains nonnegative gene programs and $H \in \mathbb{R}_{\geq 0}^{k \times n}$ contains sample loadings. Additionally, let $y \in \mathbb{R}_{>0}^{n}$ denote patient survival times, $\delta \in \{0,1\}^n$ the censoring indicators ($\delta_i = 1$ if the event is observed), and $\beta \in \mathbb{R}^k$ the Cox regression coefficients. Survival outcomes are modeled through a Cox proportional hazards model with factor scores $Z = W^\top X \in \mathbb{R}^{k \times n}$ and linear predictor $Z^\top \beta$.

The DeSurv Model

DeSurv integrates NMF with penalized Cox regression to identify gene programs associated with survival outcomes. The method operates in a semi-supervised framework: the reconstruction loss penalizes deviation from the observed expression data, while the survival term directs gene programs toward outcome-relevant structure. The parameter $\alpha \in [0, 1)$ controls the relative contribution of each objective.

The joint objective is

$$\mathcal{L}(W, H, \beta) = (1 - \alpha)\, \mathcal{L}_{\mathrm{NMF}}(W, H) - \alpha\, \mathcal{L}_{\mathrm{Cox}}(W, \beta), \tag{1}$$

where $\mathcal{L}_{\mathrm{NMF}}(W, H)$ is the NMF reconstruction error (including an ℓ_2 penalty on H) and $\mathcal{L}_{\mathrm{Cox}}(W, \beta)$ is the elastic-net penalized partial log-likelihood. A critical architectural choice is where survival supervision enters the factorization. Because factor scores are defined as $Z = W^\top X$, the Cox partial likelihood $\mathcal{L}_{\mathrm{Cox}}$ is a direct function of W and β, and the survival gradient acts explicitly on the gene program matrix W. The sample-level loadings H enter only through the reconstruction term and receive no survival gradient, preserving their interpretation as mixture coefficients[11,34]. When $\alpha = 0$, DeSurv reduces to standard unsupervised NMF. The Cox component can also accommodate additional sample-level covariates, such as tumor stage or grade, during optimization.

Optimization alternates updates for H, W, and β (Algorithm 1). The reconstruction term is minimized using multiplicative updates that preserve nonnegativity; the supervision term enters W's gradient as a Cox partial-likelihood derivative; and β is updated by Coxnet coordinate descent on the current factor scores $Z = W^\top X$ with elastic-net penalty parameters (λ, ξ). Although the joint problem is non-convex, the alternating updates converge to a stationary point under mild regularity conditions (SI Appendix, Theorem 1). Closed-form update equations, the convergence proof, and BO control parameters are given in SI Appendix, Sections 1–3.

Algorithm 1 DeSurv block-coordinate descent

Input: Expression matrix $X \in \mathbb{R}_{\geq 0}^{p \times n}$, survival (y, δ), supervision strength $\alpha \in [0, 1)$, rank k, regularization (λ, ξ), tolerance ϵ

Output: $W \in \mathbb{R}_{\geq 0}^{p \times k}$, $H \in \mathbb{R}_{\geq 0}^{k \times n}$, $\beta \in \mathbb{R}^k$

1: Initialize $W^{(0)}$ via consensus-based seeding (SI Appendix, Section 6); set $H^{(0)}, \beta^{(0)}$
2: **repeat**
3: $H \leftarrow$ multiplicative-update minimizer of $\mathcal{L}_{\text{NMF}}(W, \cdot)$
4: $W \leftarrow$ multiplicative update with combined gradient $\nabla\mathcal{L} = (1-\alpha)\nabla_W \mathcal{L}_{\text{NMF}} - \alpha \nabla_W \mathcal{L}_{\text{Cox}}(W, \beta)$
5: $\beta \leftarrow$ Coxnet coordinate descent on $Z = W^\top X$ with penalty (λ, ξ)
6: **until** $|\mathcal{L}^{(t)} - \mathcal{L}^{(t-1)}|/|\mathcal{L}^{(t-1)}| < \epsilon$

Hyperparameter selection and cross-validation

Hyperparameters $(k, \alpha, \lambda, \xi, n_{\text{top}})$ were selected by maximizing the cross-validated C-index using Bayesian optimization (BO)[35,36], which finds near-optimal configurations using far fewer objective evaluations than grid or random search over mixed integer–continuous hyperparameter spaces, with final rank k chosen by the one-standard-error rule (the smallest k whose predicted performance lies within one standard error of the maximum). All model selection was performed entirely within training data using nested cross-validation; no validation cohort data were used during any stage of tuning. Each fold was trained using multiple random initializations, and fold-level performance was defined as the average C-index across initializations. For stability, we used a consensus-based initialization for the final model, aggregating multiple DeSurv runs into a gene-gene co-occurrence matrix and constructing an initialization W_0 from the resulting clusters (SI Appendix, Section 6). Before validation, each column of W was truncated to its BO-selected number of top genes (details in SI Appendix, Section 6), denoted $\widetilde{W}$. Columns of W were normalized using the same convention applied during training (column ℓ_2-norm scaling; SI Appendix, Section 6), and the resulting truncated matrix $\widetilde{W}$ was held fixed for all validation cohorts. In PDAC, BO selected $n_{\text{top}} = 270$ top genes per factor.

Because the goal is prognostic factorization, DeSurv selects k using the criterion that directly measures prognostic performance (cross-validated C-index) rather than the unsupervised diagnostics commonly used for NMF rank selection[12,24]. In the PDAC training data, standard NMF diagnostics (reconstruction residuals, cophenetic correlation, mean silhouette width) yielded inconsistent guidance across candidate ranks, a known limitation of unsupervised rank-selection criteria that can disagree on the same dataset (SI Appendix, Fig. S5). Cross-validated concordance varied across candidate ranks but plateaued from $k = 3$ through $k = 12$ (SI Appendix, Fig. S4). The one-standard-error rule applied to joint BO over rank and supervision strength accordingly selected the parsimonious $k = 3$ at $\alpha = 0.334$.

External validation cohorts were scored by projecting new datasets onto the learned programs via $Z = \widetilde{W}^\top X_{\text{new}}$, and survival associations were evaluated using C-index, hazard ratios, and log-rank statistics. Because the survival-supervised information lives in W rather than H, scoring is a pure matrix projection: no retraining, no sample-level nonnegative least-squares estimation on the new data, and no validation survival information is required at the scoring step. This is a property not shared by methods that route survival supervision through H, which require sample-loading re-estimation on each new cohort[28,29]. Validation survival outcomes were therefore used only for downstream performance evaluation, not for model training, scoring, or hyperparameter tuning. Reported external hazard ratios and log-rank statistics therefore reflect purely out-of-sample generalization. Further training and validation details appear in SI Appendix, Part I.

Simulation studies

Simulation studies were conducted to assess recovery of prognostic latent structure and survival prediction under controlled conditions. Gene expression data were generated from a nonnegative factor model $X = WH$, where gene loadings W comprised three gene classes: marker genes, background genes, and noise genes. Marker genes were simulated to load strongly on a single factor and weakly on others, background genes to load strongly across all factors, and noise genes to have uniformly low loadings. Specifically, marker gene primary loadings were drawn from $\text{Gamma}(\text{shape} = 3, \text{rate} = 0.8)$ and cross-loadings onto other factors from $\text{Gamma}(1, 20)$; background gene loadings from $\text{Gamma}(2, 1)$ across all factors; and noise gene loadings from $\text{Gamma}(1, 20)$. Sample-level factor activities H were drawn from $\text{Gamma}(2, 2)$. Each dataset comprised $G = 3{,}000$ genes, $N = 200$ samples, and $K = 3$ factors, with 150 marker genes per factor and 500 shared background genes. Cox regression coefficients were $\beta = (2, 0, 0)^\top$ in the primary scenario, $\beta = 0$ in the null scenario, and $\beta = (2, 0, 0)^\top$ in the mixed scenario where survival risk was driven by 300 genes comprising the 150 factor-1 markers and 150 randomly sampled background genes, creating partial overlap between the prognostic signal and variance-dominant outcome-neutral programs.

Survival times were generated from an exponential distribution in which risk depended on marker gene expression through $X^T W_{\text{true}}$, where W_{true} retained marker gene loadings for their corresponding factors and was zero otherwise; censoring times were generated independently from an exponential distribution. Each dataset was analyzed using DeSurv and standard NMF followed by Cox regression on inferred factors, both tuned using cross-validated concordance index via Bayesian optimization. Performance was summarized across repeated simulation replicates. We considered three simulation scenarios: a primary scenario where prognostic programs explain low variance relative to outcome-neutral signals, a null scenario with no survival signal, and a mixed scenario where prognostic and variance-dominant programs partially overlap.

Real-world datasets

We analyzed publicly available RNA sequencing (RNA-seq) and microarray cohorts of PDAC with corresponding overall survival outcomes. Gene expression matrices were analyzed on a log scale: RNA-seq cohorts as $\log_2(\text{TPM} + 1)$; microarray cohorts in platform-normalized log-scale intensities. For each training cohort separately, genes were ranked by mean expression and by variance independently; the top 3,000 genes with the lowest sum of these two ranks were retained, selecting genes that are both highly expressed and highly variable; the intersection across training cohorts yielded 1,970 genes used for all analyses. Validation datasets were restricted to this same gene set. Survival times and censoring indicators were taken from the associated clinical annotations. Of the seven PDAC cohorts we considered, two were used for training (TCGA and CPTAC) and five were used for external validation (Dijk, Moffitt, PACA-AU array, PACA-AU RNA-seq, and Puleo). To harmonize differences in scale across cohorts, filtered gene expression data were within-subject rank transformed before model training (ranks are inherently nonnegative, preserving NMF compatibility). More details about the datasets can be found in SI Appendix, Section 7.

Per-cohort participant flow is summarized in SI Appendix, Table S1. Of 321 pooled training-cohort samples (TCGA-PAAD, $n = 181$; CPTAC-3, $n = 140$), 48 were excluded for missing survival time, missing event indicator, or quality-control failure, yielding 273 analytic training samples (139 events). Of 979 pooled validation-cohort samples, 363 were excluded for non-PDAC histology, non-tumor tissue, or missing survival annotation, yielding 616 analytic validation samples (414 events) across five cohorts: Dijk ($n = 90$, 81 events), Moffitt GEO array ($n = 123$, 83), PACA-AU array ($n = 63$, 38), PACA-AU RNA-seq ($n = 52$, 31), and Puleo array ($n = 288$, 181). With 139 events and 3 factor-score covariates (one per DeSurv factor at $k = 3$), the Cox-stage events-per-variable ratio is 46, well above the conventional 10-EPV threshold, supporting estimation stability.

Data, Materials, and Software Availability

All datasets used in this study are publicly available and de-identified; no Institutional Review Board (IRB) approval was required. Training data were obtained from the Genomic Data Commons: TCGA-PAAD[37] and CPTAC-3[38]. Validation cohorts were obtained from ArrayExpress (Dijk, E-MTAB-6830[39]; Puleo, E-MTAB-6134[40]), the Gene Expression Omnibus (GEO; Moffitt, GSE71729[41]), and the International Cancer Genome Consortium (ICGC) Data Portal (PACA-AU array and RNA-seq; European Genome–phenome Archive (EGA) study EGAS00001000154[42]). Cohort details, sample sizes after quality control, and patient-level metadata used are summarized in SI Appendix, Section 7. Expression data and molecular classifications (PurIST, DeCAF) for the validation cohorts were originally compiled for the DeCAF study[5].

All analyses were performed in R (version 4.6.0; R Core Team) using the following key packages: DeSurv (v1.0.1; this work), `NMF`[34], `glmnet`[43], `survival`, `ggplot2`, and `cowplot`. Bayesian optimization for hyperparameter selection was implemented within DeSurv. An R package implementing DeSurv is available at github.com/rashidlab/DeSurv (archived at Zenodo: DOI `10.5281/zenodo.20150929`). Code, processed data, and the exact session-info file (R + package versions used to render this manuscript) are available at github.com/rashidlab/DeSurv-paper (archived at Zenodo: DOI `10.5281/zenodo.20150946`).

Data and Code Availability

The DeSurv R package is available at https://github.com/rashidlab/DeSurv (archived at Zenodo, DOI `10.5281/zenodo.20150929`). Reproducibility code, processed data, and session-info for this manuscript are available at https://github.com/rashidlab/DeSurv-paper (archived at Zenodo, DOI `10.5281/zenodo.20150946`). All training and validation cohorts are publicly available: TCGA-PAAD and CPTAC-3 from the Genomic Data Commons; Dijk (E-MTAB-6830) and Puleo (E-MTAB-6134) from ArrayExpress; Moffitt (GSE71729) from the Gene Expression Omnibus; and PACA-AU (EGA study EGAS00001000154) from the ICGC Data Portal. The treated bulk-tissue cohorts used in the exploratory translational analysis (RASH-ACCEPT and the Linehan FOLFIRINOX$_{\pm}$CCR2-inhibitor series) are restricted-access clinical-trial datasets that are not publicly deposited; raw data are available from the respective study investigators under a data-use agreement, subject to the governing trial consents and institutional approvals. To preserve reproducibility without redistributing restricted data, the derived per-cohort summary statistics reported in the main text and SI Appendix are provided in the reproducibility repository (`results/treated_cohort_stats.rds`, generated by `code/11_treated_cohort_analysis.R`), so that every treated-cohort value can be inspected and verified.

Acknowledgements

This work was supported by the National Cancer Institute grants U01 CA274298, P50 CA257911, T32 CA106209, and R01 CA288145, and by the Department of Defense Peer Reviewed Cancer Research Program (HT9425241103100121). We thank the patients and research teams who contributed samples and clinical data to the public cohorts used here (TCGA, CPTAC, Dijk, Moffitt, Puleo, PACA-AU).

Inclusion and Diversity. The patient cohorts analyzed in this study were curated by the original investigators or consortia (TCGA, CPTAC, ICGC, and the original Moffitt, Dijk, and Puleo studies) and our access was limited to the de-identified expression and survival data they provide; we did not stratify analyses by sex, race, or ethnicity because consistent demographic annotations were not available across all cohorts. Sex and other demographic variables that could modify the prognostic relevance of the gene programs identified here remain important directions for follow-up work as more comprehensively annotated cohorts become available.

Author Contributions

Conceptualization: A.M.Y., N.U.R. Methodology: A.M.Y., A.Y., D.L., N.U.R. Software: A.M.Y. Formal analysis: A.M.Y. Investigation: A.M.Y. Data curation: A.M.Y., X.L.P. Visualization: A.M.Y. Writing – original draft: A.M.Y., N.U.R. Writing – review and editing: A.M.Y., A.Y., X.L.P., J.J.Y., D.L., N.U.R. Resources (PDAC domain expertise and biological interpretation): X.L.P., J.J.Y. Supervision: N.U.R. Funding acquisition: A.M.Y., N.U.R. All authors reviewed and approved the final manuscript.

Competing Interests

N.U.R. and J.J.Y. are named as inventors on US patents US20220127676A1 and US20250066856A1, which relate to pancreatic cancer subtype classification (establishment of Basal and Classical subtypes and their prediction) derived from prior work and were not used directly in this study. The other authors declare no competing interests.

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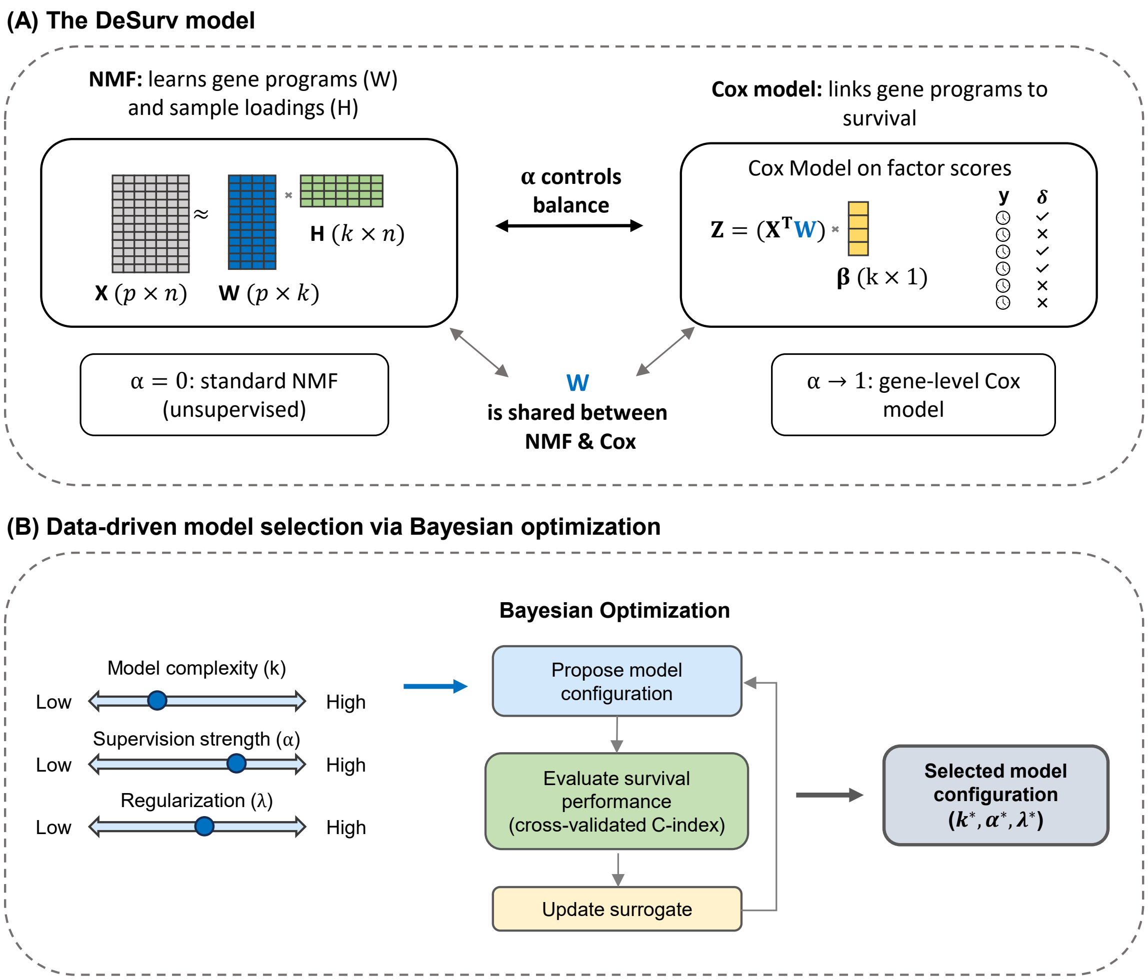

Figure 1: Overview of the DeSurv framework. (A) DeSurv jointly optimizes NMF reconstruction ($X \approx WH$) and Cox proportional hazards likelihood. The gene program matrix W is shared between objectives: factor scores $Z = W^\top X$ serve as covariates in the Cox model with coefficients β. A supervision parameter $\alpha \in [0, 1)$ controls the balance between reconstruction fidelity ($\alpha = 0$, standard NMF) and survival prediction (α close to 1). (B) The factorization rank (k), supervision strength (α), and regularization (λ) are selected via Bayesian optimization over cross-validated concordance index.

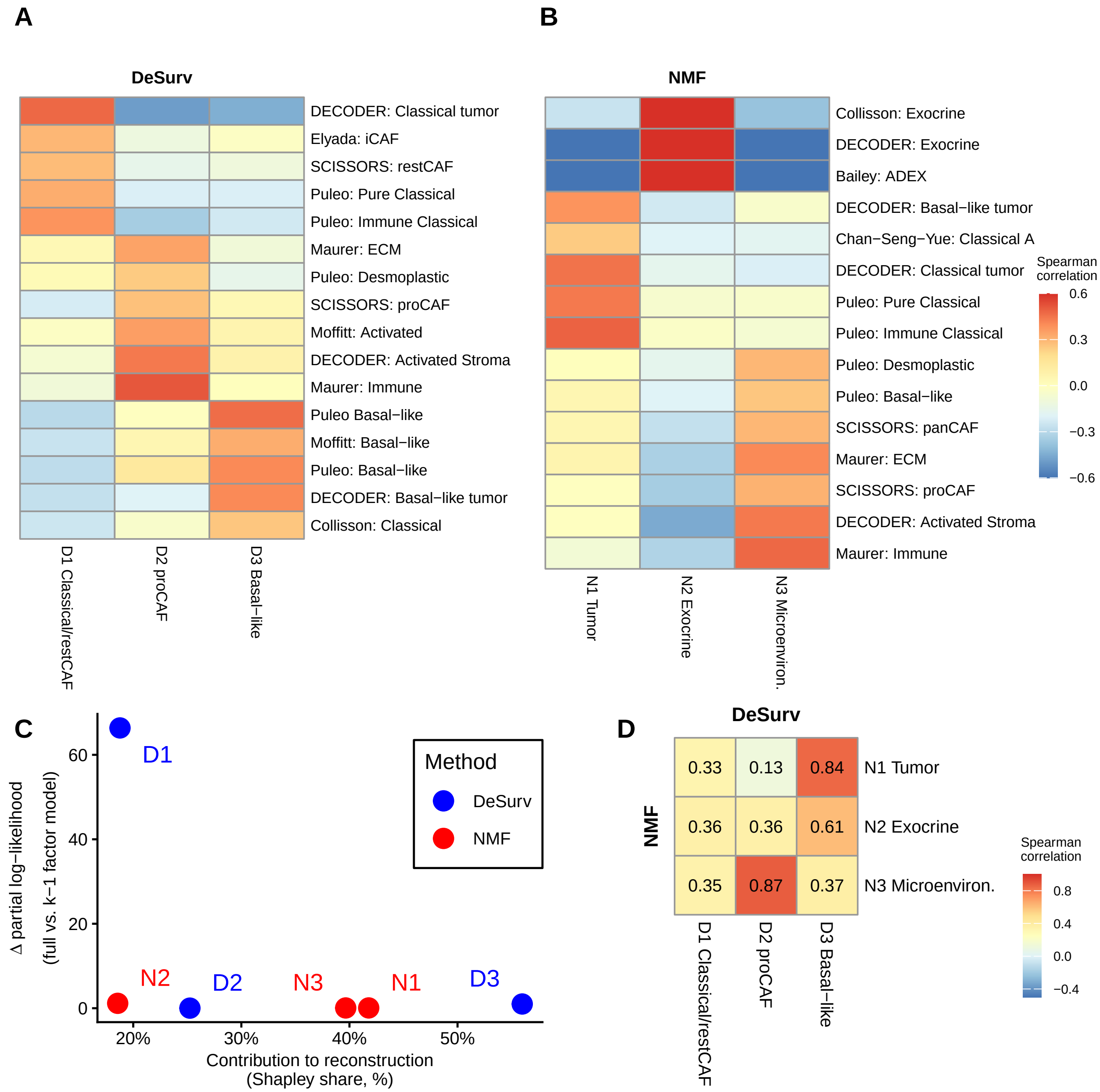

Figure 2: Survival supervision produces different factor structures in PDAC (TCGA + CPTAC training cohort, $n = 273$ patients, 139 events). (A–B) Spearman correlations between factor gene rankings (top 50 genes per factor) and established PDAC gene programs for DeSurv (A) and standard NMF (B) at $k = 3$. Only $|r| > 0.2$ shown. DeSurv concentrates survival signal into D1 (Classical tumor + restCAF-like stroma) while suppressing exocrine variation; standard NMF devotes one factor to exocrine signals. (C) Per-factor contribution to reconstruction of the expression matrix (normalized reconstruction Shapley share, which partitions and sums to 100%) versus survival contribution (Δ Cox partial log-likelihood upon factor removal). DeSurv concentrates survival signal into D1 while NMF's largest-reconstruction factors — including the exocrine-associated factor N2 — contribute negligibly to survival. (D) Pairwise correspondence between NMF and DeSurv factors, measured by Spearman rank correlation of W-matrix gene loadings over all genes.

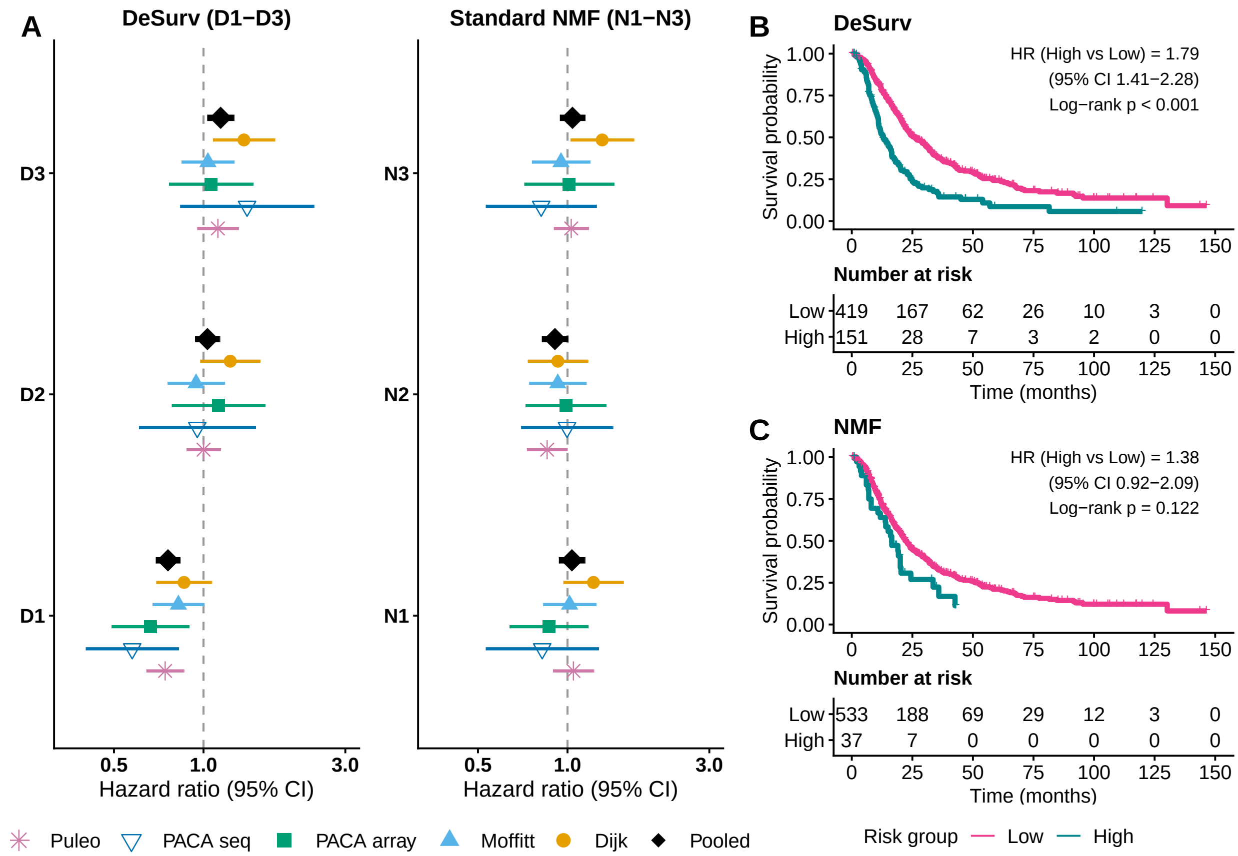

Figure 3: Survival-aligned programs generalize across independent PDAC cohorts (pooled $n = 616$ patients across 5 cohorts: Dijk $n = 90$; Moffitt GEO array $n = 123$; PACA-AU array $n = 63$; PACA-AU RNA-seq $n = 52$; Puleo array $n = 288$). (A) Forest plot of univariate hazard ratios (95% CIs) from separate Cox models for each factor, showing each factor's marginal survival association in each validation cohort. DeSurv factor D1 shows a consistent protective association across all five cohorts (pooled HR = 0.76), while no standard NMF factor achieves a comparable pooled effect. Error bars are 95% confidence intervals; black diamonds denote inverse-variance-weighted pooled estimates. (B–C) Kaplan–Meier curves for pooled external validation samples stratified by the combined multivariate linear predictor ($\hat{\beta}_1 Z_1 + \hat{\beta}_2 Z_2 + \hat{\beta}_3 Z_3$, where coefficients are learned from training data), dichotomized at a cross-validated optimal cutpoint: DeSurv (B) and standard NMF (C).

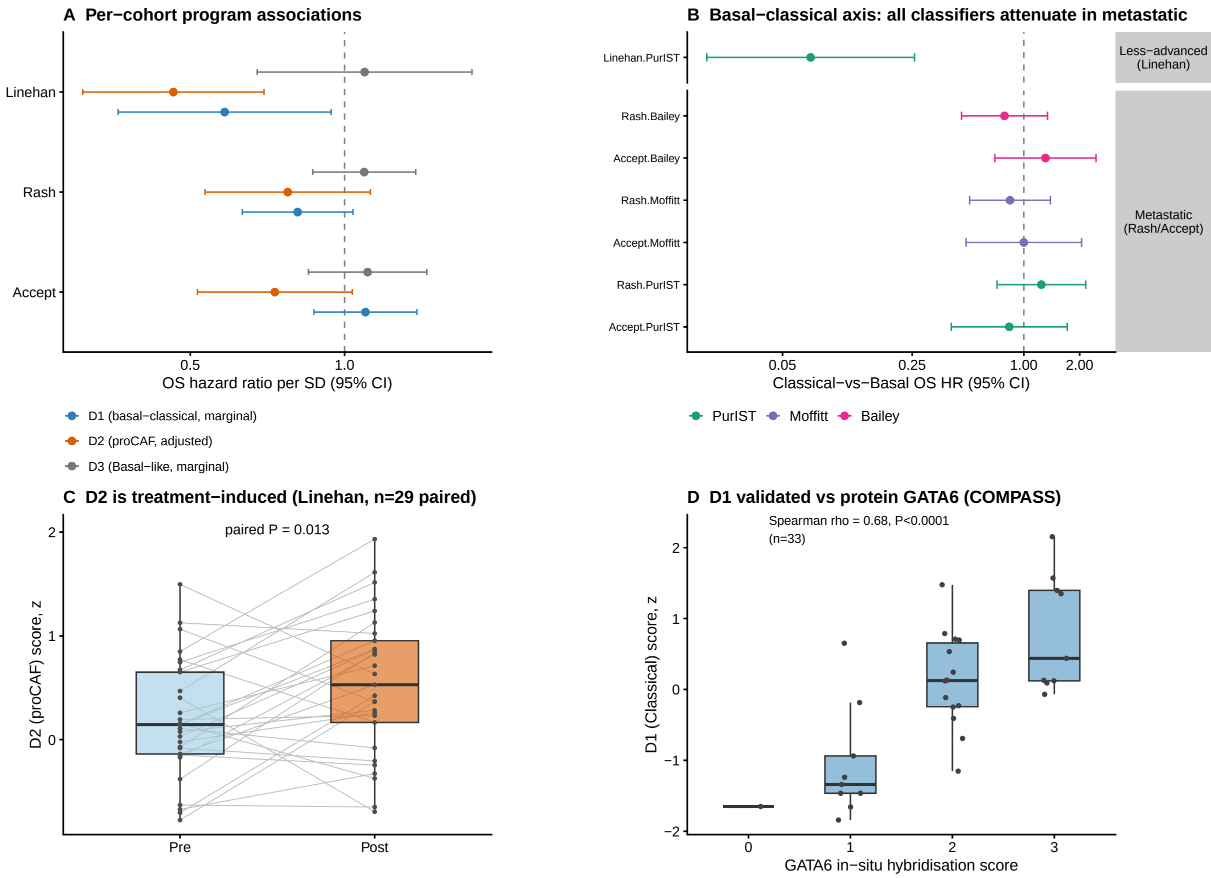

Figure 4: Survival-supervised programs reveal a context-dependent prognostic shift in treated PDAC. Trained DeSurv programs were projected without retraining onto three bulk treated cohorts (Linehan, borderline-resectable/locally-advanced, $n = 66$; Rash, metastatic, $n = 85$; Accept, metastatic, $n = 70$) and analysed per cohort. (A) Per-cohort overall-survival hazard ratios per SD for each program (D1 and D3 marginal; D2 PurIST + DeCAF-adjusted); D2 (proCAF) is protective in all three cohorts, D1 (basal–classical) is prognostic only in the less-advanced cohort, and D3 is null on overall survival. (B) The basal–classical prognostic axis is setting-dependent and concordant across measures: Classical-vs-Basal overall-survival hazard ratios for the PurIST, Moffitt, and Bailey classifiers in less-advanced versus metastatic disease, all attenuating together in metastatic disease. (C) The proCAF program D2 increases from pre- to post-treatment in paired biopsies (paired Wilcoxon test). (D) Orthogonal validation: the D1 score versus protein-level GATA6 in-situ hybridisation (Spearman correlation).

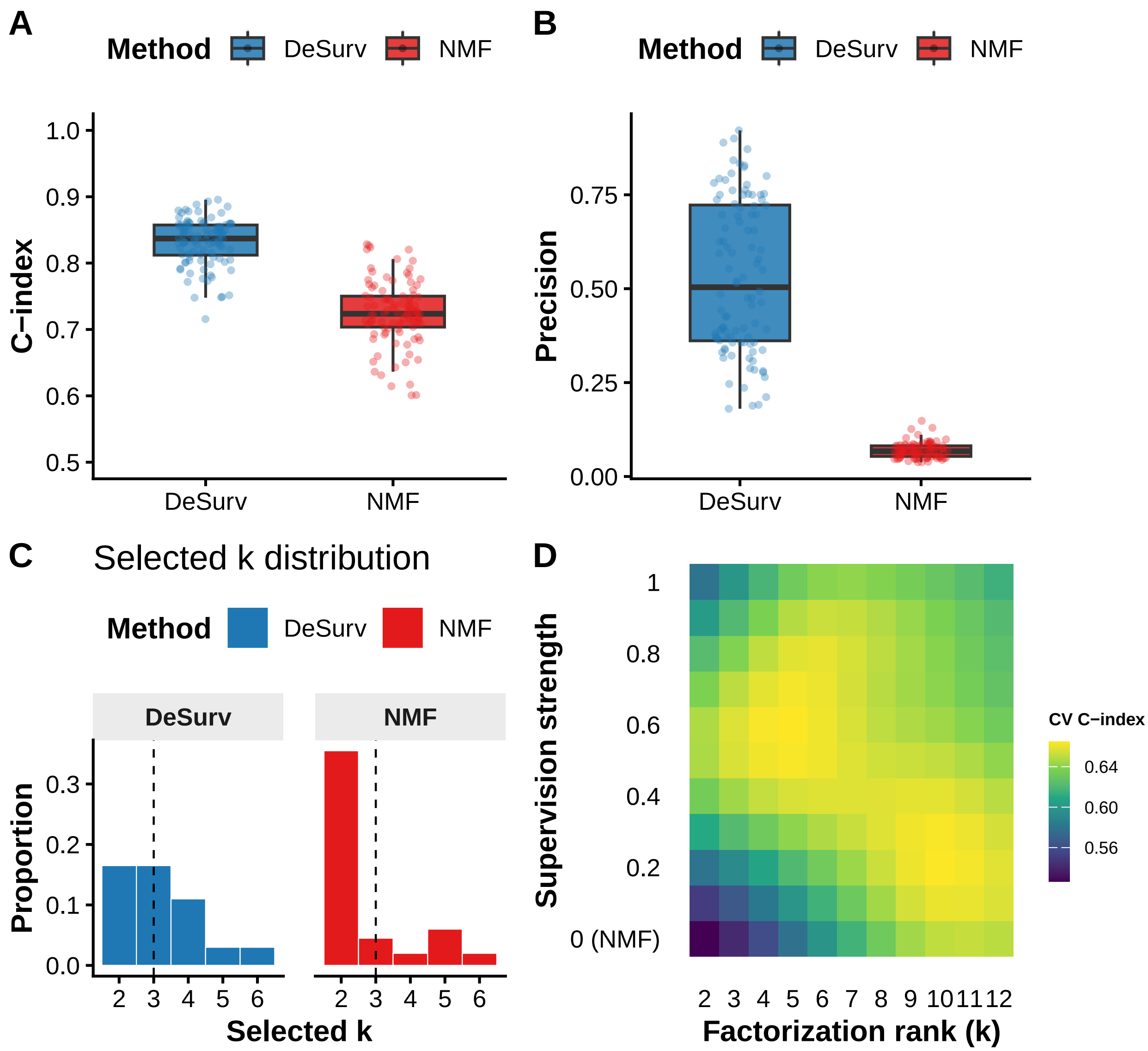

Figure 5: Survival supervision improves recovery of true prognostic gene programs in controlled simulations. (A) Test-set concordance index across 100 simulation replicates ($p = 3{,}000$ genes, $n = 200$ samples, true $k = 3$) comparing DeSurv to standard NMF ($\alpha = 0$); paired Wilcoxon $P < 0.001$. (B) Precision (fraction of a factor's top-weighted genes, defined in main text, overlapping a true prognostic program) across replicates. In (A) and (B), boxes show median and interquartile range (IQR); whiskers extend to 1.5×IQR; points beyond are individual replicates. The y-axis in (A) is truncated at 0.5 because C-index values below 0.5 indicate worse-than-random concordance. (C) Distribution of selected k values across the same 100 replicates for DeSurv versus standard NMF. (D) Gaussian-process–predicted mean cross-validated C-index from Bayesian optimization over factorization rank (k) and supervision strength (α) in PDAC training data (TCGA + CPTAC), showing the broad plateau spanning $k = 3$–12 from which the one-standard-error rule selects $k = 3$ (Methods).